FEDERAL SPECIFICATION SHEET

FEDERAL SPECIFICATION

DRYING AND CONDITIONING TUMBLERS, LAUNDRY, COMMERCIAL:

PASS THROUGH TYPE, END LOADING STYLE, REVERSIBLE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

(This specification forms a part of the latest issue of Federal Specification 00-D-1297).

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the detail requirements for reversible, pass through, end loading tumblers.

1.2 Classification. The tumblers shall be of the size and kind of heat as specified (see 6.2).

Size: 200 or 400 (dry-weight pounds)
Kind of heat: steam, gas, butane, propane, oil, or gas/oil

2. APPLICABLE DOCUMENTS

2.1 Specifications. The following Federal Specifications of the issue in effect on date of invitation for bids or request for proposal, forms a part of the specification to the extent specified herein:

00-D-1297 - Drying and Conditioning Tumblers, Laundry, Commercial: General Specification For

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FSC 3510

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. REQUIREMENTS

.1 Design and construction. The tumblers shall be in accordance with le I and shall be equipped to tilt the cylinder for loading and unloading. tumblers shall be capable of receiving and drying and conditioning tightly ked damp laundry and then discharging same without causing "roping", or ngling". Arrangement shall also be provided for non-circulation of the ted air. The machine shall be designed to use not less than 70 percent side air and shall be provided with cool-down controls.

TABLE I - Design requirements

e		Evaporation	Capacity	
imum weight acity unds)	Type of heat	Water evaporated per minute, minimum (pounds)	Conditioned dry weight per hour, (pounds)	Drying dry weight per hour, (pounds)
)	steam	5	1800	600
)	gas, butane, oil, propane, gas/oil	10	2400	800
)	steam	5.5	3000	1200
)	gas, butane, oil, propane, gas/oil	15	4800	1600

.1.1 Housing. The housing containing the cylinder shall tilt a minimum 15 degrees above and below the horizontal. The housing shall be so con- ucted to prevent placement of objects under the tilting portion of the hine.

.1.2 Tilting mechanism. The tilting mechanism shall be motorized gears hydraulic or air powered pistons. The tilting assembly shall be so con- ucted and pivoted that the tumbling portion of the operating cycle will be formed with the cylinder in a horizontal position. Automatic operation shall vide for stopping the rotation of cylinder housing assembly at "loading", erating", and "unloading" position for proper operation.

.1.3 Doors. The tumbler shall be equipped with two doors (front and rear) they shall not be smaller than 48 inches in diameter. Each door shall be emi or automatically operating type, mounted at the front of the unit. The r shall either move on trolleys in a track or be of the overhead sliding folding type. The door shall be interlocked with the tilting controls to vent opening until the cylinder is in the "load" or "unload" position.

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3.1.4 Cylinder. The cylinder shall have a minimum diameter of 66 inches. The cylinder shall be open at each end and be fabricated of perforated metal. An alternative fabricating method is to use galvanized steel woven wire mesh sheets in lieu of the perforated sheets. The cylinder shall be suitable reinforced with not less than six lifting ribs and a frame and shall be supported by rollers equipped with polyurethane threads. Rotation of the cylinder shall be provided by either motor driven support rollers or belt(s) which encircle the cylinder.

3.1.5 Moisture retention control. Unless otherwise specified (see 6.2), the tumblers shall be provided with a moisture retention control. The control shall be adjustable from 15 to 45 percent moisture retention. When the preset moisture content has been reached, the control shall automatically shut off the heat.

3.1.6 Lint collector. The lint collector shall be a separate corrosion resistant component capable of filtering tumbler exhaust prior to its release to outside atmosphere. The lint collector shall be compatible with the exhaust requirements of the tumbler and shall be connected to the tumbler by adequately sized ductwork. Pressure drop through the lint collector shall not exceed 1/2 inch of water pressure. The lint collector shall not support combustion and shall be either a wet or dry type as specified (see 6.2). Dry type collectors shall be equipped with a fire extinguishing system. Exterior surface temperatures of the lint collector shall conform to the requirement for the exterior surface temperature of the tumbler when the collector is installed in the interior of any building. The intended location of the lint collector shall be as specified (see 6.2).

3.1.7 Surface temperature. During maximum temperature performance, 85 percent of the tumbler's exterior surface shall have temperatures not exceeding 115 F. and 15 percent shall not exceed 200 F. The ambient temperature shall be 80 F. for testing.

3.1.8 Burner. The burner shall be equipped with safety devices as specified (see 3.6.1.28 of 00-D-1297) and shall be provided with controls to maintain an adjustable preset stack operating temperature variable to 225 F. maximum. The machine shall be equipped with controls to provide an optional maximum inlet temperature of 350 F. for processing durable press materials and a dial temperature gauge to indicate inlet temperature. When specified (see 6.2), the tumbler shall be equipped with a manual transfer facility to allow for the use of propane gas or oil as the heat media in the event of discontinuance of the normal gas supply. An exhaust system shall be furnished in accordance with the manufacturer's current practice. The exhaust system furnished shall be compatible with the style machine specified.

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3.1.9 Insulation. Insulation used to comply with the exterior surface temperature requirement (see 4.8.5 of 00-D-1297) shall be fire-resistant and shall be resistant to and installed to prevent settling, falling off or separating from the open shell. Metal panels shall be used to enclose insulation attached to the machine exterior, except the doors.

4. QUALITY ASSURANCE PROVISIONS

4.1 Visual examination. In addition to the general classification of defects, the following shall apply:

Examine	Defect	Classification Major	Minor
Construction and workmanship, specific	Housing construction to prevent placement of objects under tilting portion	X	
Electrical assembly	Moisture retention control, missing	X	

4.2 Test items, functional testing. During functional testing, the following shall be checked for compliance with the specified requirements:

(a) Automatic "load", "operate", and "unload" tilting cycles.
(b) Door and tilt interlock to prevent automatic opening of door until the cylinder is in the load or unload position.
(c) Operation of tumbler allowed only in horizontal cylinder position.
(d) Surface temperature.

5. PREPARATION FOR DELIVERY (See 00-D-1297)

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) The size (see 1.2).
(b) The kind of heat (see 1.2).
(c) When a moisture retention control is not required (see 3.1.5).
(d) Whether the lint collector is to be located inside or outside the building (see 3.1.6).
(e) Whether lint collector is to be wet or dry type (see 3.1.6).
(f) When manual transfer facility for other type fuels is required (see 3.1.8).

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Army - GL

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